

Predicting Short-Term Customer Spend (30-Day CLV) in Retail Using Regression

1. Business Background

Retailers care not just about what a customer spent today, but about how much they are likely to spend in the near future.

In this project, you will build a regression model that predicts short-term customer value: the total amount a customer is expected to spend in a fixed future window (e.g., next 30 days), based on their past behaviour and characteristics.

You will work in a team of 4, follow fast sprints, and find your own dataset from public sources.

2. Problem Definition

2.1 Business Problem

For each active customer, we want to estimate "How much will this customer spend in the next 30 days?" This is sometimes called short-term CLV (Customer Lifetime Value) over a defined horizon.

2.2 Machine Learning Problem

- **Type:** Supervised learning – regression
 - **Input (X):** Features describing each customer up to a cutoff date, such as:
 - Historical transaction behavior (past spend, number of orders, recency).
 - Product/category preferences.
 - Channel usage (online vs in-store, web vs mobile).
 - Basic customer attributes (if available): region, segment, loyalty tier, etc.
 - **Target (y):** Continuous numeric value:
 - Total monetary amount the customer spends in the next 30 days after the cutoff date.
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3. Goals and Success Criteria

3.1 Primary Goals

- 3.1.1 Modeling goal:** Build at least one regression model that predicts 30-day customer spend with reasonable accuracy on held-out customers.
 - 3.1.2 Engineering goal:** Package the model so it can be called via an API-style function/endpoint and build a simple front-end UI to query predictions.
 - 3.1.3 Communication goal** Explain clearly:
 - How you defined "short-term spend" (the target).
 - What features you used.
 - How well the model works, in a short final presentation with a working demo.
 - 3.1.4 Quality / Success Metrics:** Use appropriate regression metrics like MAE, RMSE, R^2 or any other relevant to domain.
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4. Data Requirements (Students Must Choose the Dataset)

You must find a public dataset that supports the problem. Some common *sources*: open data portals, Kaggle, UCI, etc.

Your dataset must satisfy:

- 4.1 **Retail context**: Dataset must represent a retail or e-commerce *business* (Online shop, supermarket, fashion retailer, marketplace, etc.) It should *contain* transaction-level data, where each row is a purchase event.
- 4.2 **Customer identification**: There must be a customer identifier (cust_id/user_id). Each customer should have multiple transactions over time, *not* just one.
- 4.3 **Timestamps**: Transactions must have a date/time field (InvoiceDate, order_date). This is needed to separate past behaviour from future spend.
- 4.4 **Monetary measure**: Each transaction must have enough information to compute its monetary value, for example: A field like TotalAmount, or Quantity and UnitPrice so you can compute transaction_value.
- 4.5 **Size and structure**: Ideally at least 5,000 customers or 20,000+ transactions. A mix of numeric and categorical fields like Product category, country, channel, etc.
- 4.6 **Data usability**: Dataset must be publicly available and legally usable for educational purposes. It should be in a usable format (CSV, JSON, etc.).

5. Sprint Breakdown and Expected Outcomes

You will work in three sprints within 6 total hours.

5.1 Sprint 1 (1 hour): System Design

Objective: Understand the problem, choose a dataset, and design the overall solution.

Key tasks:

- 5.1.1 Restate the problem in your own words
- 5.1.2 Dataset scouting and choice: Identify 1–2 candidate datasets and verify they are retail/e-commerce, they have customer_id, timestamps, and transaction values (direct or computable). There are multiple purchases per customer.
- 5.1.3 High-level architecture: Language & libraries. How/where the model will be served? How the UI will call the model (e.g., local API, function call)?

Deliverables:

- Short written problem summary (business + ML).
 - Chosen dataset + quick justification.
 - Defined horizon (e.g., 30 days) and planned cutoff date logic.
 - High-level system diagram.
 - Rough allocation of roles within the team.
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5.2 Sprint 2 (4 hours): Data Selection, EDA and Model training

Objective: Prepare customer-level data, define features and 30-day future spend target, and explore.

Key tasks:

- 5.2.1 Load and clean transactions
- 5.2.2 Create customer-level features like RFM, Category, Customer attributes
- 5.2.3 Define target variable
- 5.2.4 EDA

Deliverables:

- Customer-level dataset: One row per customer, Features from past behavior., Target future_spend_30d.
 - Simple EDA summary showing: Descriptive stats and plots. Initial hypotheses.
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Model Selection, Training, Evaluation, Packaging

Objective: Train and evaluate regression models, pick a final one, and prepare it for inference.

Key tasks:

- 5.2.5 Data splitting
- 5.2.6 Baseline model
- 5.2.7 Model selection, training and hyper-parameter tuning
- 5.2.8 Model evaluation
- 5.2.9 Interpretability (Optional)
- 5.2.10 Packaging

Deliverables:

- Trained and saved final model and preprocessing.
 - Training code and evaluation results.
 - Comparison of baseline vs models.
 - Short write-up of key insights.
 - A callable inference interface ready for integration.
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5.3 Sprint 3 (1 hour): Front-End UI and Endpoint Integration

Objective: Build a simple UI and connect it to your model so users can see predictions.

Key tasks:

- 5.3.1 Design UI inputs
- 5.3.2 Implement UI
- 5.3.3 Integrate UI with model
- 5.3.4 Error handling & UX

Deliverables:

- Working UI that accepts customer-like feature values and returns and displays a model prediction for short-term spend.
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Final Presentation

Objective: Present your complete solution and demonstrate the working system.

Key topics to cover:

- 5.3.5 Problem Overview
- 5.3.6 Data & Target
- 5.3.7 Features & Modeling
- 5.3.8 System & Demo
- 5.3.9 Reflection

Deliverables:

- Slide deck (or simple structured presentation).
 - Live or recorded demo of the running system.
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6. Evaluation Criteria

Teams can be evaluated on:

6.1 Problem Understanding

- Clear explanation of business value and ML framing.
- Correct definition of the target (future spend) and input features (past behavior).

6.2 Data & Modeling

- Sensible dataset choice and cleaning.
- Thoughtful feature engineering.
- Proper use of train/test split.
- Baseline comparison and improvement.

6.3 Engineering & Integration

- Clean, organized code.
- Model packaging and inference function.
- Functional UI with working integration.

6.4 Communication & Presentation

- Clear storyline: problem → data → model → results → demo.
 - Ability to answer basic questions about choices and trade-offs.
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7. Optional Extensions (If Time Allows)

If you finish early, you can explore:

- Different prediction horizons (e.g., 60 or 90 days).
- Segment-specific models (e.g., separate models by country or channel).